

Trainings ▾

General Information

Since the introduction of the K38 engines, the design of the cylinder liner counterbore was changed. Older engines do **not** contain a counterbore ring, but have a thick flange liner. Newer engines have a counterbore for thick flange cylinder liners and contain a counterbore ring.

All K50 engines contain a counterbore for thick flange cylinder liners and contain counterbore rings.

The liner protrusion for the K38 and K50 engines was changed. The liner protrusion **must** be within 0.152 mm to 0.203 mm [0.006 in to 0.008 in]. The depth of the counterbore on present blocks was changed to obtain the correct protrusion.

Coolant passage machining has been changed from 1 in NPTF to straight bored holes with an o-ring connection. Two 3/8 -16 tapped holes were added at each access hole for securing the new connection. There are three coolant access passages on each side of the block. Some holes have been relocated (see illustrations for plugs and dowels).

The front and rear coolant holes are used for the aftercooler water supply on all KV engines with side mounted aftercoolers.

The vee cover plates that attach to the block on the K38 were changed. Previous engines had two cover plates that were the same size. Present engines require one small, and three large cover plates.

The vee cover plates were changed on the K50 cylinder block. The number of cover plates remains at five, but the location of the smaller plate was changed.

The cylinder block casting for K2000E and K1800E engines, starting with engine serial number 33123861, has been modified from the original K50 cylinder block at the oil transfer cover and filter head oil inlet mounting pad to reduce oil flow restriction and increase the flow area. A different bolt pattern is used for the oil transfer cover. Older style K50 cylinder blocks can **not** be used for K2000E or K1800E engines.

K2000E and K1800E engines have two blank hand hole covers on each engine. The covers have clearance machined on the top edge to clear the oil transfer connection. The covers are to be installed in the locations below the oil transfer connection and lubricating oil filter head when blank covers are used in those locations. If a hand hole cover with a breather, oil filler tube, or dipstick is installed in the location below the oil transfer connection or lubricating oil filter head, the hand hole cover is to be installed in any other location requiring a blank cover. There will not be an interference between the oil transfer connection and the hand hole covers having oil filler tubes, breathers, or dipsticks as the cover height is 180 mm [7.09 in] on these covers, rather than 191 mm [7.52 in] on blank covers.

The cylinder block was changed to allow the installation of a longer cylinder head capscrew. Previous engines **must** have capscrews that are 171 mm [6.73 in] in length. Present engines **must** have capscrews that are 203 mm [8.00 in] in length. The short capscrews are available for service. These capscrews are cadmium plated (silver or gold colored). The short capscrews **must** be torqued to 353 N•m [260 ft-lb]. If the short capscrews have been coated with lubrite and are black in color, they **must** be replaced. The 203 mm [8 in] capscrews are **only** available with the lubrite coating. A torque of 555 N•m [409 ft-lb] **must** be applied to these capscrews. K50 engines have **always** had the 203 mm [8.00 in] capscrews.

Two capscrews have been used for the main bearing caps. One capscrew is black and has a 12-point head. The other capscrew is dark silver or gray and has a 6-point head. The capscrews can be installed on the same main cap, and the same torque specification can be applied.

The cylinder blocks for the K38 and K50 engines were revised when the KTTA38 and KTTA50 engines were introduced. These engines require an increased number of ports in the block coolant cavity for the aftercooler coolant inlet tubing. The blocks with the increased number of ports were produced for all K38 and K50 engines. The cylinder head surface of these blocks was revised to allow for the attachment of turbocharger oil drain tubes with an o-ring and capscrew. Previous blocks contained holes with pipe threads for the connection of the oil drain tubes.

Previous K38 and K50 cylinder blocks contained a hole at the right bank rear and the left bank front that allowed the attachment of the crankcase breathers. The crankcase breathers are mounted on the intake manifold on revised block engines. A plate and a gasket are available to plug the old breather holes when the revised design of the breather is installed. Revised cylinder blocks contain holes that are near the location of the previous breather holes. The holes in the revised blocks are for the connection of the turbocharger oil drain hose.

A further revision in the cylinder block was the bore for the water pump idler gear shaft. The bore is larger and deeper so that a new shaft can be installed. This new shaft provides a greater area of contact with the block. The new shaft requires a change in the counterbore in the front gear housing for the flange on the shaft. A special service tool kit is available to machine an old block and gear housing for the new shaft.

In November 1993, beginning with engine serial number 33124631, K38 and K50 engine blocks had extra machining in two locations (except K2000E and K1800E which was carried out in January 1992) to reduce lube oil restrictions.

For a short period of time, the bores for both camshaft idler gear shafts were machined the same as the new water pump idler gear shaft bore. The following engine serial numbers require idler shaft, Part Number 3176836, in the camshaft idler locations.

33110658	33110708	33110726
33110684	33110709	33110727
33110689	33110710	33110730
33110688	33110711	33110731
33110690	33110712	33110732
33110695	33110714	33110733

33110696	33110715	33110736
33110698	33110716	33110738
33110702	33110719	33110740
33110704	33110720	33110742
33110705	33110723	
33110706	33110724	

Present cylinder blocks do **not** contain threads for pipe plugs in some locations. These pipe plugs were replaced with expandable rivet type plugs (AVSEAL®). The AVSEAL® **must** be removed to clean the part properly. A service tool for the removal and installation of these plugs is available, Part Number 3823154. The hole for the AVSEAL® is smaller than the hole for a pipe plug. The hole **must** be reamed and tapped to install a pipe plug until the new tools are available.

All crankshaft oil seals on the K38 and K50 engines are the Teflon® lay-down lip (scroll) type. Many other shaft seals are the Teflon® lay-down lip type. The Teflon® lay-down lip type seal does **not** contain a spring on the back of the sealing lip. The sealing lip is a thin stiff piece of Teflon®. Some Teflon® type seals contain a second sealing lip that performs as a dust lip. These seals are required in a severe operating environment, such as a high dust environment.

Teflon® seals **must** be dry before installation. Do **not** lubricate the seal lip or the shaft. After the first few turns of the shaft, a thin film of Teflon® is transferred from the seal lip to the shaft. The seal will leak if the shaft is **not** clean and dry because the Teflon® can **not** be transferred.

If the seal is **not** the Teflon® type, lubricate both the sealing lip and the shaft with engine oil.

All oil seals **must** be installed in one of two ways; even with the cast surface, or even with the bottom of the entry chamfer on the seal bore.

Engines that have a wet type flywheel housing **must** have a double-lip type seal for the rear of the crankshaft. This type of seal is required to prevent torque converter oil from entering the engine.

There are four lifting brackets on the K38 and K50 engine that bolt to the cylinder block. These lifting brackets are designed to hold the weight of an assembled engine **only**.

⚠ WARNING ⚠

Inspect the engine lifting brackets for damage or cracks. Do not attempt to lift the engine if any cracks or damage is visible. Using damaged lifting equipment can cause serious person injury and/or property damage.

⚠ WARNING ⚠

The engine lifting brackets must be properly and securely attached to the engine before proceeding with the engine removal. Failure to do so can result in equipment damage, severe personal injury or death.

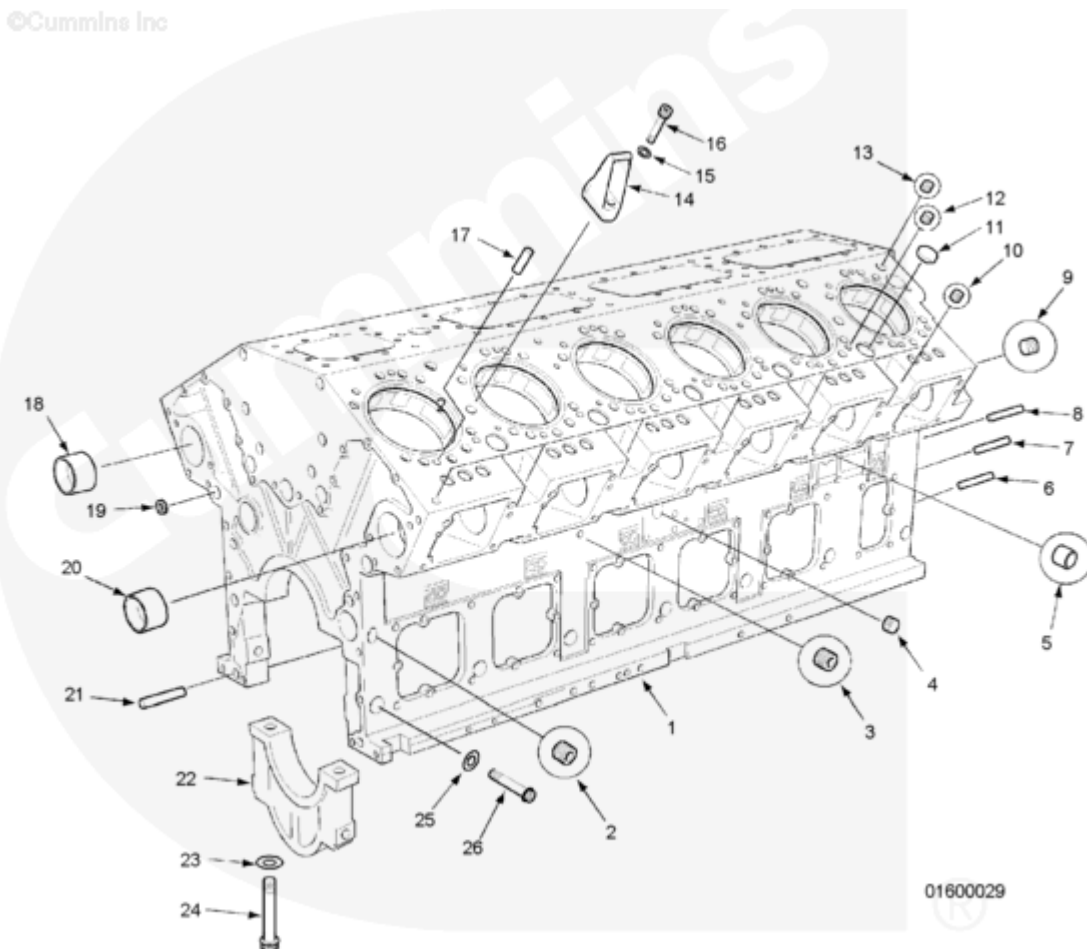
⚠ WARNING ⚠

The engine lifting brackets are designed to lift the engine safely. Do not lift the engine, gearbox, if fitted, and main generator as an assembly. Exceeding the weight limit of the engine lifting brackets can result in equipment damage, severe personal injury or death.

The lifting brackets on the block are capable of lifting an engine that has a sea (raw) water pump and a marine heat exchanger attached.

Before installation of the lifting brackets, a room temperature vulcanizing sealant **must** be applied. The sealant prevents entry of water and subsequent corrosion of the capscrew. Corrosion of the capscrew can cause the part to break when the lifting bracket is used.

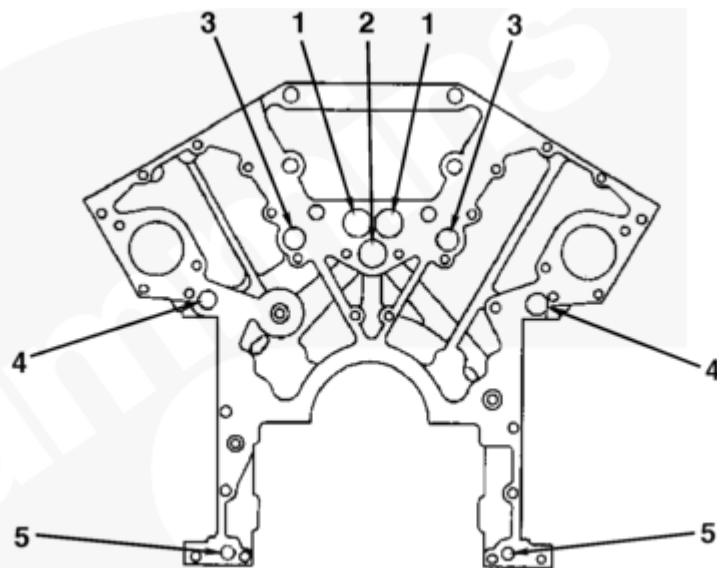
Exploded View



1. Cylinder block
2. Pipe plug
3. Pipe plug

4. Pipe plug
5. Ring dowel
6. Dowel pin
7. Dowel pin
8. Dowel
9. Pipe plug
10. Pipe plug
11. Expansion plug
12. Expansion plug
13. Pipe plug
14. Lifting bracket
15. Plain washer
16. Hexagon head capscrew
17. Groove pin
18. Bushing
19. Expansion plug
20. Bushing
21. Dowel pin
22. Main bearing cap
23. Plain washer
24. Hexagon head capscrew
25. Plain washer
26. Hexagon head capscrew.

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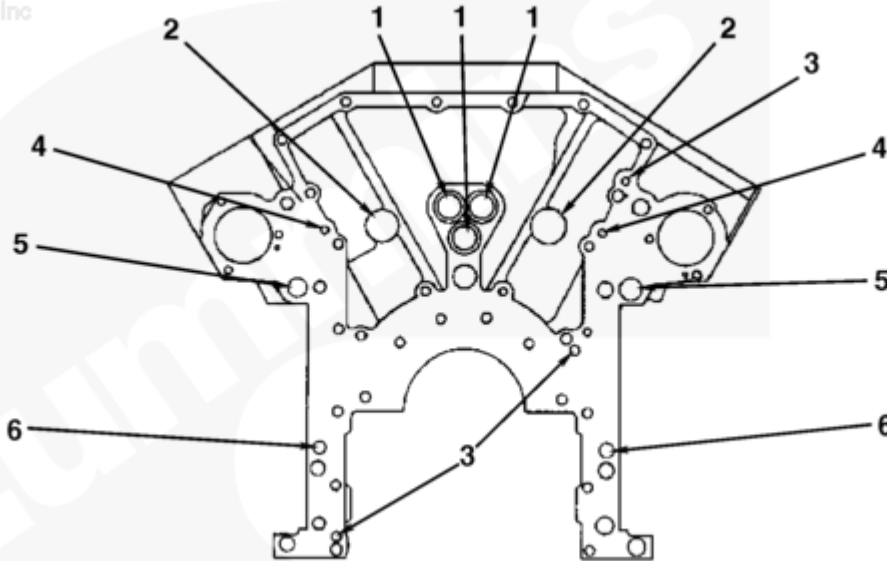
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Plugs and Dowels - K38 and K50 (Front)

1. Main oil rifle cup plug
2. No plug (This **must** be open to provide oil to the front gear housing and cover.)
3. Coolant passage drilling cup plug, (Newer blocks; most K38 and K50 engines do **not** have this hole. Older K38 engines have a cup plug or a pipe plug).
4. Camshaft oil rifle cup plug

5. Front gear housing and cover dowel.

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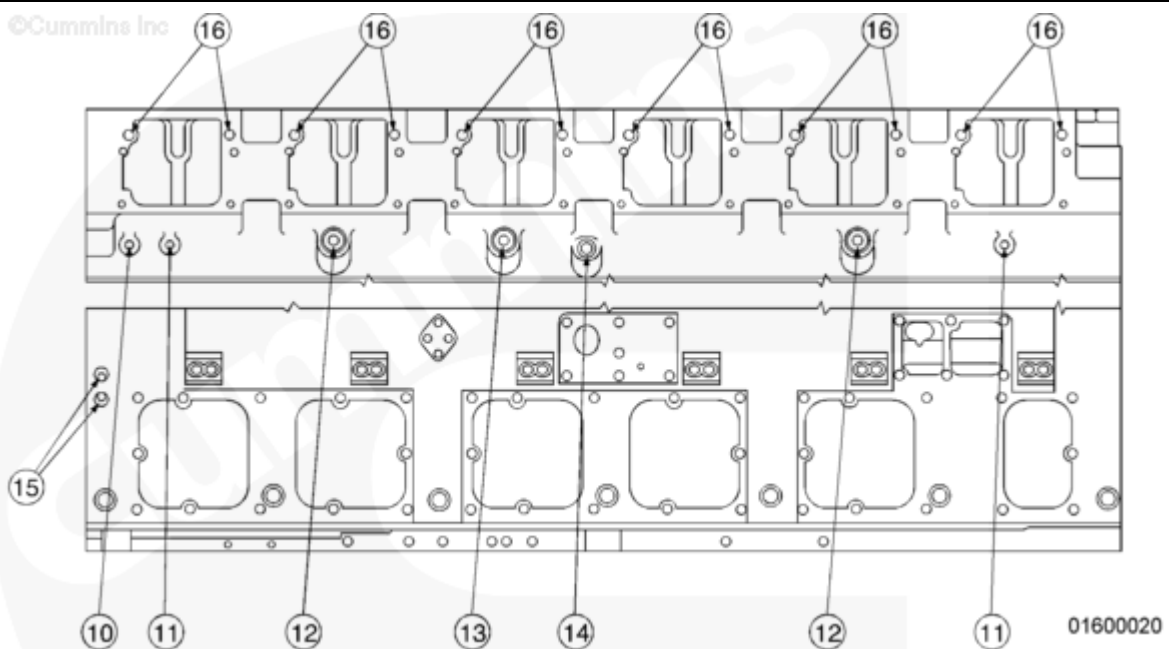


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Plugs and Dowels - K38 and K50 (Rear)

1. Main oil rifle cup plug
2. Coolant passage drilling cup plug, (Newer blocks; most K38 and K50 engines do **not** have this hole. Older K38 engines have a cup plug or a pipe plug).
3. Rear seal housing or rear gear drive dowel - lower unit
4. Plate and seal (**Only** if the engine has rear seal housing; **must** be open if engine has rear gear drive lower unit to provide oil to the rear gear drive lower and upper units. The left bank (LB) and right (RB) plates are different. Some blocks do **not** contain these holes.)
5. Camshaft oil rifle cup plug
6. Flywheel housing dowel.

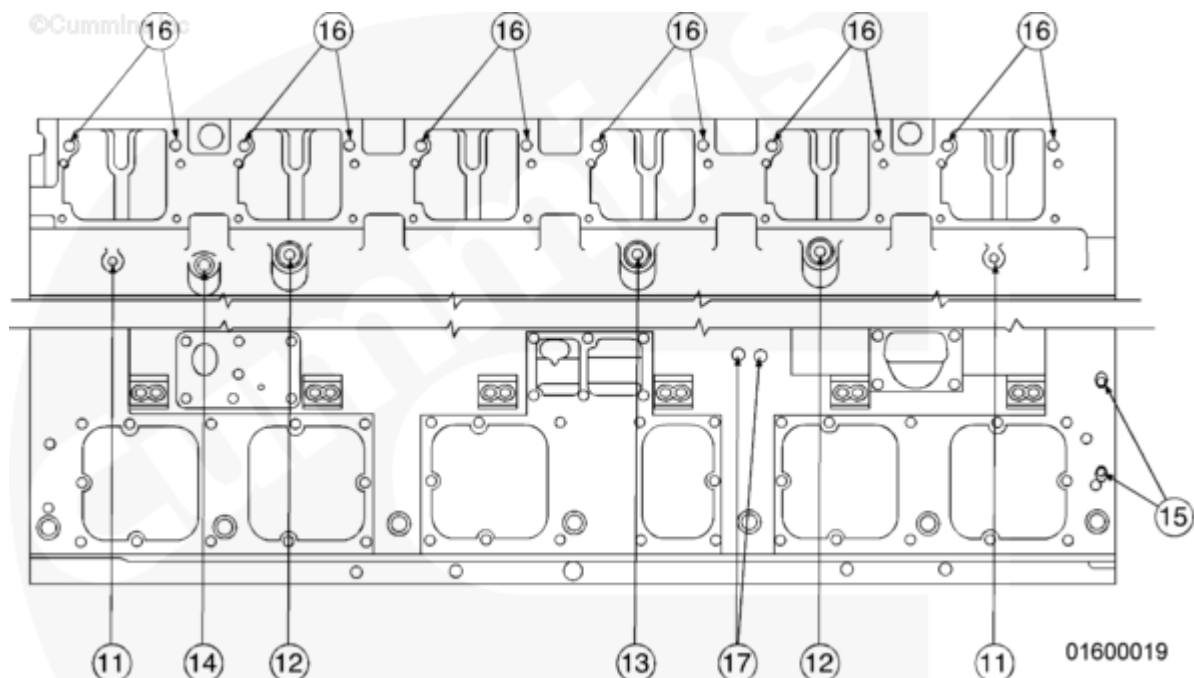
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Plugs and Dowels - K38 (Left Bank)

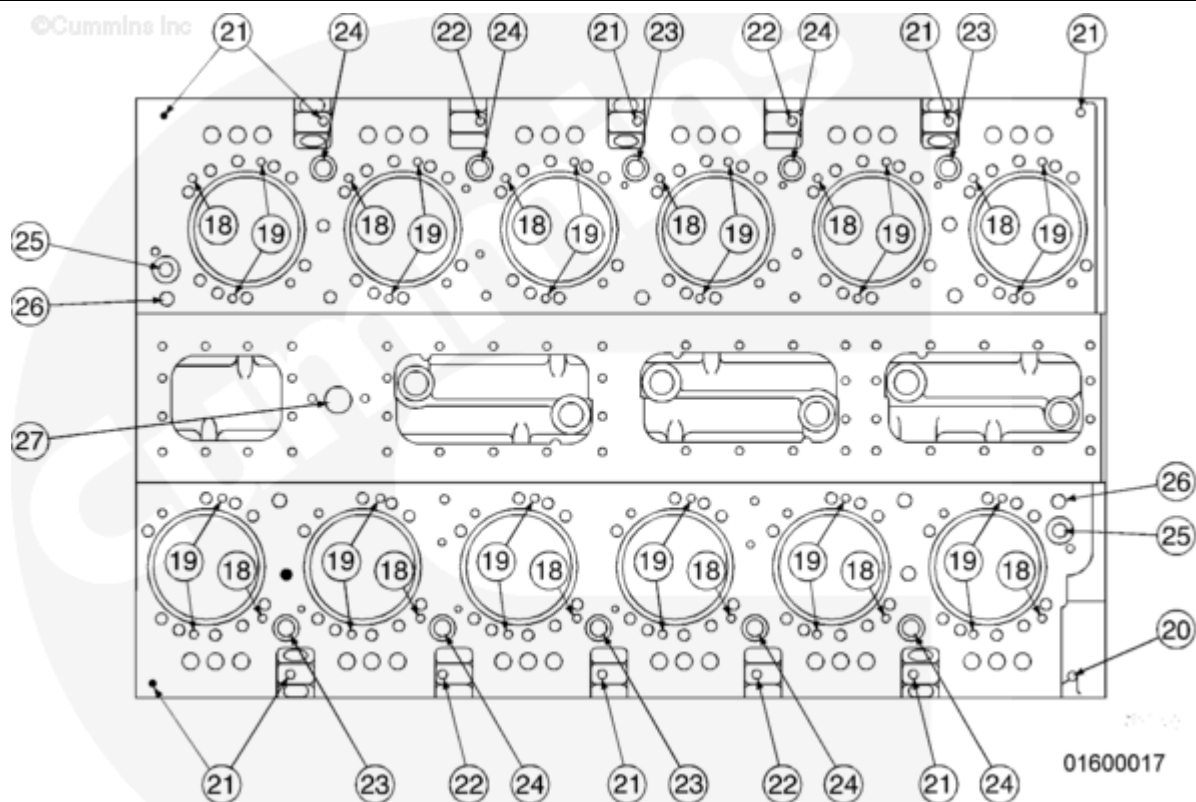
10. Air compressor water inlet hose fitting, if equipped with an air compressor; pipe plug if no air compressor.
11. Cylinder block coolant draincock.
12. Coolant block passage plug (KTTA and KTA engines with outboard aftercoolers have a fitting for the aftercooler coolant inlet hose).
13. Block coolant passage plug (This can be a fitting for a vehicle heater return hose on some applications; **not** installed on older blocks).
14. Main oil rifle pipe plug (On engines with Step Timing Control (STC), a fitting for the STC control valve oil inlet hose is to be installed).
15. AVSEAL® or pipe plug (oil drilling for idler shaft bores).
16. Cam follower shaft ring dowel (no need to remove for cleaning).



Plugs and Dowels - K38 (Right Bank)

11. Cylinder block coolant draincock.
12. Coolant block passage pipe plug (KTTA and KTA engines with outboard aftercoolers have a fitting for the aftercooler coolant inlet hose.)
13. Block coolant passage pipe plug (This can be a fitting for a vehicle heater return hose on some applications; **not** installed on older blocks.)
14. Main oil rifle pipe plug (on engines with STC, a fitting for the STC control valve oil inlet hose is to be installed).
15. AVSEAL® or pipe plug (oil drilling for idler shaft bores).
16. Ring dowel, cam follower shaft ring (no need to remove for cleaning).

17. Fuel tube or hose fitting (The hole nearest to the front of the block is for fuel pressure. The hole nearest to the rear of the block is for fuel drain.)



Plugs and Dowels - K38 (Top)

18. Overhead oil rifle restriction plug.

19. Cylinder head groove pin.

20. Oil drilling pipe plug (Common for vehicle oil pressure gauge fitting.)

21. Oil drilling AVSEAL® or pipe plug.

22. Oil drilling pipe plug (flange type oil inlet hose for turbocharger in some engines).

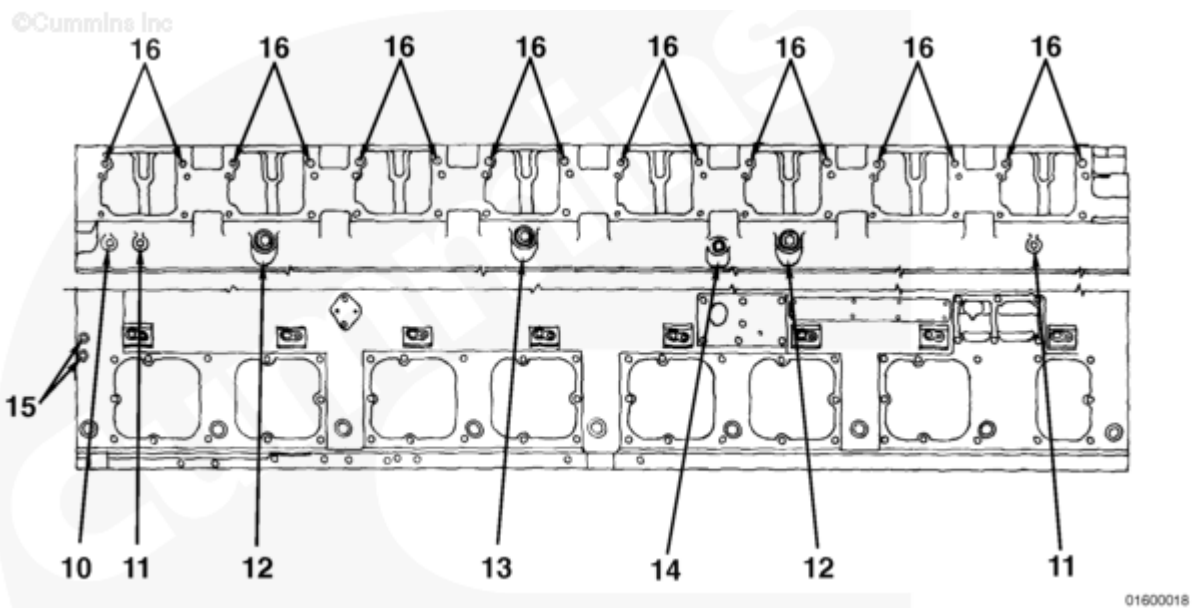
23. Crankcase drilling cup plug (flange type oil drain hose for turbocharger in some engines).

24. Crankcase drilling cup plug.

25. Crankcase drilling cup plug (flange type turbocharger drain in some engines; **always** a cup plug on KTTA engines).

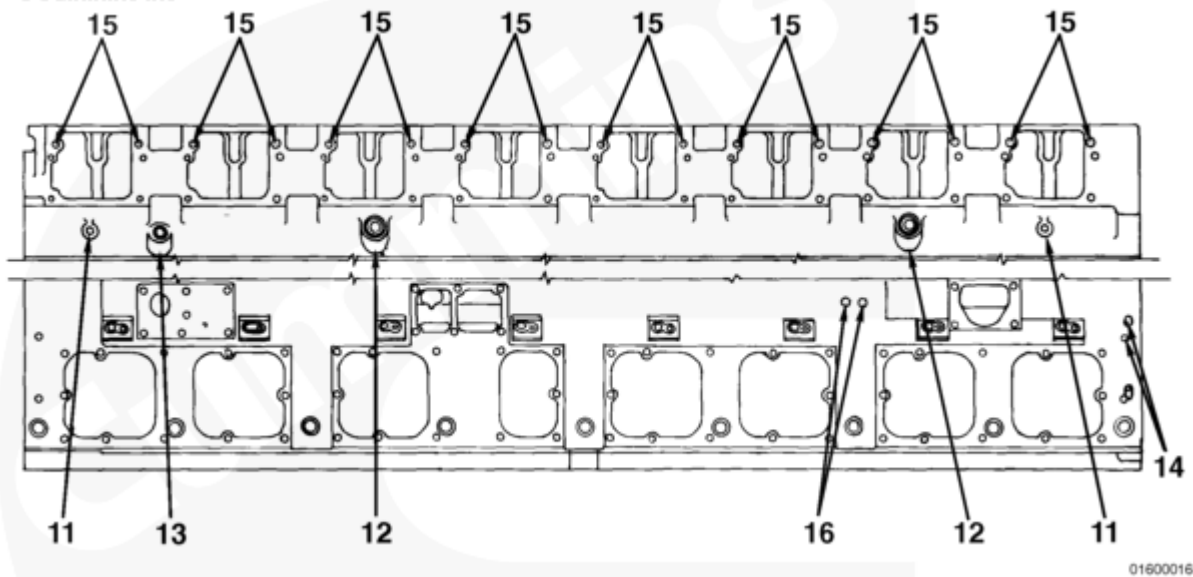
26. Crankcase drilling cup plug (On newer engines this is a straight threaded tapped hole for turbocharger support bracket. Older engines contain a hole with pipe threads for oil drain hose fitting for turbocharger.)

27. Crankcase drilling cup plug (On engines with outboard aftercoolers, a gasket and plate are required. On engines with center mount aftercooler, it is the mounting for the aftercooler water inlet connection).



Plugs and Dowels - K50 (Left Bank)

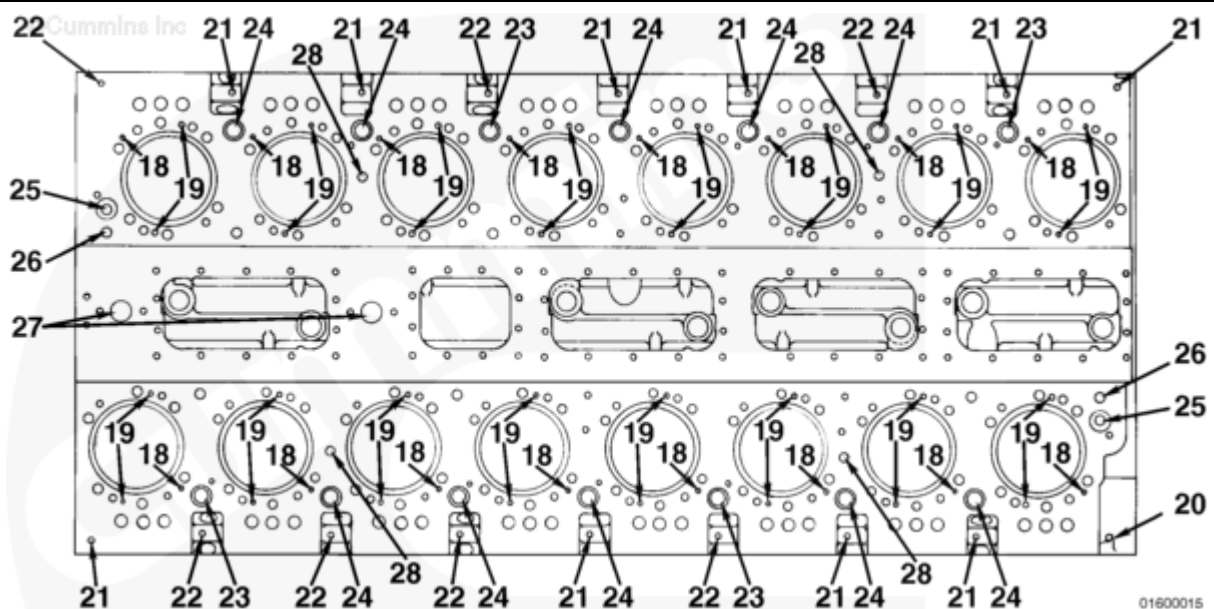
10. Air compressor water inlet hose fitting, if equipped with an air compressor; pipe plug if no air compressor.
11. Cylinder block coolant draincock.
12. Coolant block passage plug (KTTA and KTA engines with outboard aftercoolers have a fitting for the aftercooler coolant inlet hose.).
13. Block coolant passage plug (This can be a fitting for a vehicle heater return hose on some applications; **not** installed in older blocks.).
14. Main oil rifle pipe plug (On engines with STC, a fitting for the STC control valve oil inlet hose is to be installed.).
15. AVSEAL® or pipe plug (oil drilling for shaft bores).
16. Cam follower shaft ring dowel (no need to remove for cleaning).



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Plugs and Dowels - K50 (Right Bank)

- 11. Cylinder block coolant draincock.
- 12. Coolant block passage plug (KTTA and KTA engines with outboard aftercoolers have a fitting for the aftercooler coolant inlet hose.).
- 13. Main oil rifle pipe plug (On engines with STC, a fitting for the STC control valve oil inlet hose is to be installed.).
- 14. AVSEAL® or pipe plug (oil drilling for idler shaft bores).
- 15. Cam follower shaft ring dowel (no need to remove for cleaning).
- 16. Fuel tube or hose fitting (The hole nearest to the front of the block is for fuel pressure. The hole nearest to the rear of the block is for fuel drain).



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Plugs and Dowels - K50 (Top)

- 18. Overhead oil rifle restriction plug.

19. Cylinder head pin.
20. Oil drilling pipe plug (common for vehicle oil pressure gauge fitting).
21. Oil drilling AVSEAL® or pipe plug.
22. Oil drilling pipe plug (flange type oil inlet hose for turbocharger in some engines).
23. Crankcase drilling cup plug (flange type oil drain hose for turbocharger in some engines).
24. Crankcase drilling cup plug.
25. Crankcase drilling cup plug (flange type turbocharger drain in some engines; **always** a cup plug on KTTA engines).
26. Crankcase drilling cup plug (On newer engines, this is a straight threaded tapped hole for the turbocharger support bracket. Older engines contain a hole with pipe threads for a oil drain hose fitting for the turbocharger.).
27. Crankcase drilling cup plug (On engines with outboard aftercoolers, a gasket and plate are required. On engines with center mount aftercooler, it is the mounting for the aftercooler water inlet connection.).
28. Lifting bracket holes plug.

Clean and Inspect for Reuse

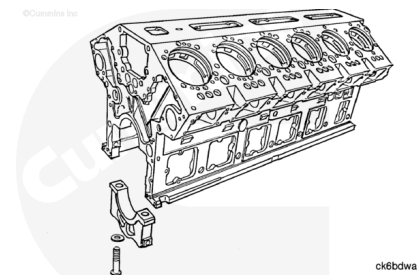
⚠ CAUTION ⚠

Use care when cleaning the engine block. Machined gasket surfaces and camshaft bushings can be damaged.

Use a scraper, a wire brush, or a Scotch-Brite™ pad to clean the heavy dirt deposits from the engine block.

- Gasket surfaces
- Mounting surfaces
- Cylinder liner counterbore ledge and the press fit areas
- Cylinder liner packing ring bore
- Top of the engine block
- Main bearing saddles and the caps
- Cup plug bores.

Use a bottle brush with a long handle to clean all the oil passages.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury. Be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a cleaning solution that will not damage the camshaft bushings or the AVSEAL® sleeves.

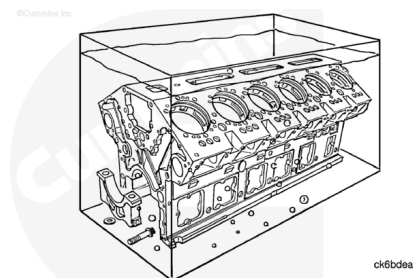
Remove the engine block from the engine stand and place the block in a cleaning tank.

Follow the instructions of the manufacturer of the cleaning tank and the manufacturer of the cleaning solution.

Note : Cummins Inc. does not recommend any specific cleaning solution.

Experience indicates that the best results are obtained using a cleaning solution that can be heated to 82°C to 93°C [180°F to 199°F].

A cleaning tank that will mix and filter the cleaning solution gives the best results.

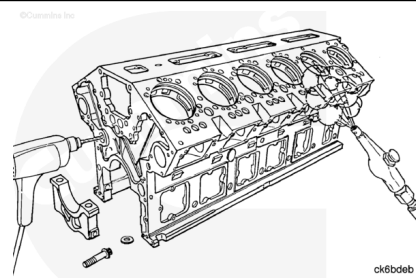


⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of engine block damage, make sure all the water is removed from the capscrew holes and the oil passages.

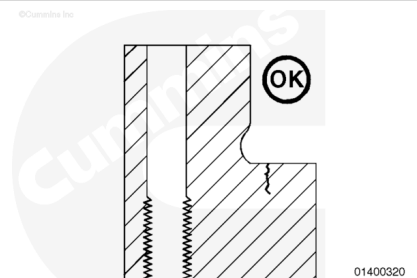
Remove the block from the cleaning tank.

Clean the engine block with a steam cleaner. Make sure all the oil passages are clean.

Use compressed air to dry the block.

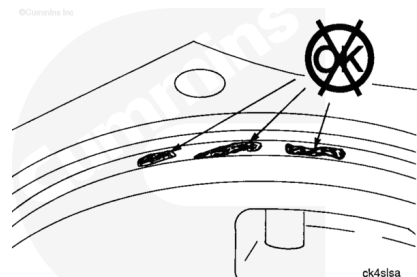
If the cylinder block is **not** going to be used immediately, apply a coating of preservative oil to prevent rust. Cover the block to prevent dirt from sticking to the oil.

Metallurgical analysis of cross sections of counterbores having circumferential cracks has revealed that the cracks are surface-initiated on the top of the counterbore ledge, and normally do **not** propagate vertically through the counterbore ledge into the coolant passage around the cylinder liner.



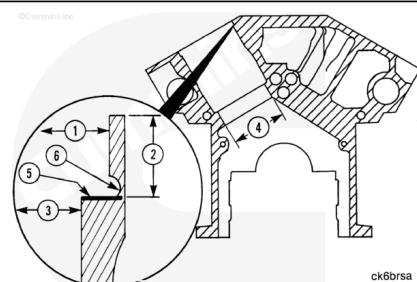
Check the liner seats of the engine block for pitting.

Pitting on the liner seat is **not** acceptable. This illustration shows an example of pitting in an unacceptable area. This block requires counterbore machining before it can be reused. Contact a Cummins® Authorized Repair Location.



⚠ CAUTION ⚠

All measurements of the cylinder block must be performed with the block positioned on a flat surface. If the block is mounted on an engine stand, distortion will cause the measurements to be wrong.



To inspect the block properly, the counterbore design **must** be determined.

Definition of numerical terms (call-outs):

- 1. Upper counterbore, inside diameter
- 2. Counterbore depth
- 3. Lower counterbore, inside diameter
- 4. Packing ring bore
- 5. Counterbore ledge
- 6. Counterbore radius.

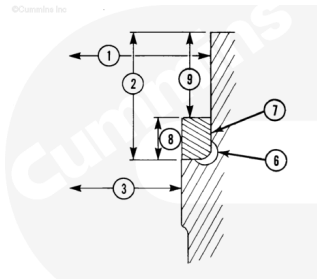
Review the following illustrations to determine the counterbore design of the block.

Note : This section applies to Counterbore Ring - Thick Deck Block Design.

Most K38 engines with an engine serial number higher than 33100193 contain this design.

All K50 engines contain this design.

The diameter (1) depends on whether standard size or oversize liners were installed.



ck6Brsb

Oversize			
	mm		in
Standard	190.28	MIN	7.492
	190.34	MAX	7.494
0.010	190.53	MIN	7.502
	190.59	MAX	7.504
0.020	190.79	MIN	7.512
	190.85	MAX	7.514
0.030	191.04	MIN	7.522
	191.10	MAX	7.524
0.040	191.30	MIN	7.532
	191.36	MAX	7.534

The depth (2) depends on whether standard size or oversize liners were installed. The depth also depends on whether the block was cut for shims.

Oversize			
	mm		in

Standard	18.24	MIN	0.718
	18.29	MAX	0.720
0.010	18.49	MIN	0.728
	18.54	MAX	0.730
0.020	18.75	MIN	0.738
	18.80	MAX	0.740
0.030	19.00	MIN	0.748
	19.05	MAX	0.750
0.040	19.25	MIN	0.758
	19.30	MAX	0.760

The depth can be a maximum of 3.38 mm [0.133 in] greater if the block was machined previously for shims and oversize liners.

(3) Lower Counterbore Inside Diameter

mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028

Lower Counterbore Inside Diameter

mm		in
181.80	MIN	7.157
181.81	MAX	7.158

Note : The cylinder block can be used again **only** if the correct press fit between the cylinder liner and dimensions (1) is obtained.

(6) This design contains a double undercut radius.

(7) This design contains a counterbore ring that functions as a load distribution washer.

(8) Counterbore ring height.

Oversize

	mm		in
Standard	5.055	MIN	0.1990
	5.067	MAX	0.1995
0.002	5.105	MIN	0.2010
	5.118	MAX	0.2015

0.004	5.156	MIN	0.2030
	5.169	MAX	0.2035

(9) This depth depends on the size of the liner. Liner protrusion **must** be 0.152 to 0.203 mm [0.006 to 0.008 in].

Oversize			
	mm		in
Standard	13.18	MIN	0.519
	13.23	MAX	0.521
0.010	13.44	MIN	0.529
	13.49	MAX	0.531
0.020	13.69	MIN	0.539
	13.74	MAX	0.541
0.030	13.94	MIN	0.549
	14.00	MAX	0.551
0.040	14.20	MIN	0.559
	14.25	MAX	0.561

If the upper counterbore is **not** within standard specifications, the block **must** be machined for oversize liners. Refer to Procedure 001-058 in Section 1.

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If the block is larger than the specification for the largest oversize, a repair sleeve can be installed.

Refer to Procedure 001-058 in Section 1.

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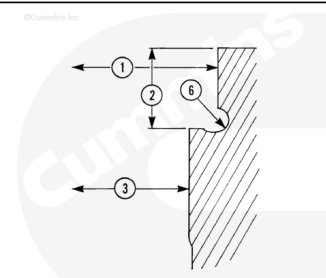
If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims. Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

Before machining the block for shims, check to see if the protrusion can be corrected by installing an oversize counterbore ring.

Note : This section applies to Thin Deck Block - Double Undercut Radius Design.

⚠ CAUTION ⚠



Do not attempt to machine blocks of this design for a counterbore ring, as it will cause the block to break.

Some K38 standby generator set engines with an engine serial number higher than 33100193 contain this design.

No K50 engines contain this design.

(1) The actual size depends on whether the standard size or the oversize liners were installed.

Oversize			
	mm		in
Standard	190.276	MIN	7.4912
	190.343	MAX	7.4938
0.01"	190.530	MIN	7.5012
	190.597	MAX	7.5038
0.02"	190.784	MIN	7.5112
	190.851	MAX	7.5138
0.03"	191.038	MIN	7.5212
	191.105	MAX	7.5238
0.04"	191.292	MIN	7.5312
	191.359	MAX	7.5338

(2) These specifications are required for a liner protrusion of 0.152 to 0.203 mm [0.006 to 0.008 in]. The protrusion specification has changed since blocks with this design were introduced. The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.

Oversize			
	mm		in
Standard	13.18	MIN	0.519
	13.23	MAX	0.521
0.010	13.44	MIN	0.529
	13.49	MAX	0.531
0.020	13.69	MIN	0.539
	13.74	MAX	0.541
0.030	13.94	MIN	0.549
	14.00	MAX	0.551
0.040	14.20	MIN	0.559

	14.25	MAX	0.561
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(3) Lower Counterbore Inside Diameter

mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028

Lower Counterbore Inside Diameter

mm		in
181.80	MIN	7.157
181.81	MAX	7.158

Note : The cylinder block can be used again **only** if the correct press fit between the cylinder liner and dimensions (1) is obtained.

(6) This design has the double undercut radius.

If the upper counterbore is **not** within standard specifications, the block **must** be machined for oversize liners. Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims. Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

If the block is larger than the specification for the largest oversize liner, a repair sleeve can be installed. Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

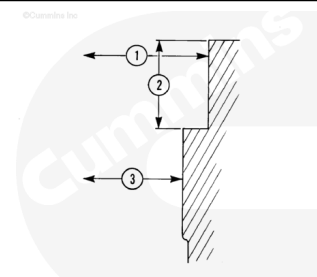
It is recommended that a block of this design be used **only** for K38 standby generator set applications.

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the block to break.

Thin Deck Block Design:

Most K38 engines with an engine serial number lower than 33100193 contain this design.



ck4brgg

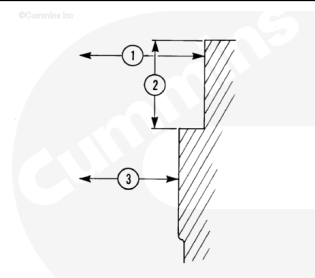
No K50 engines contain this design.

(1) The actual size depends on whether the standard size or oversize liners were installed.

Standard			
Oversize			
	mm		in
Standard	190.276	MIN	7.4912
	190.343	MAX	7.4938
0.01"	190.530	MIN	7.5012
	190.597	MAX	7.5038
0.02"	190.784	MIN	7.5112
	190.851	MAX	7.5138
0.03"	191.038	MIN	7.5212
	191.105	MAX	7.5238
0.04"	191.292	MIN	7.5312
	191.359	MAX	7.5338

Note : These specifications are those required for a liner protrusion of 0.152 mm to 0.203 mm [0.006 in to 0.008 in]. The protrusion specification has changed since blocks with this design were produced.

(2) The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.



ck4brgg

Oversize			
	mm		in
Standard	13.16	MIN	0.519
	13.23	MAX	0.521
0.010	13.41	MIN	0.529
	13.49	MAX	0.531
0.020	13.67	MIN	0.539
	13.74	MAX	0.541
0.030	13.92	MIN	0.549
	14.00	MAX	0.551
0.040	14.17	MIN	0.559
	14.25	MAX	0.561

(3) Lower Counterbore Inside Diameter		
mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028.

Lower Counterbore Inside Diameter		
mm		in
181.80	MIN	7.157
181.81	MAX	7.158

Note : The cylinder block can be used again **only** if the correct press fit between the cylinder liner and dimensions (1) is obtained.

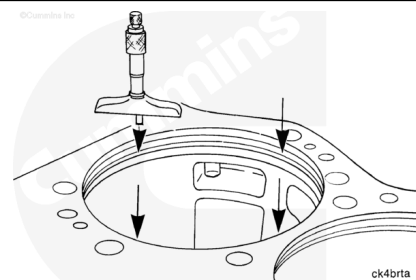
This design does **not** contain the double undercut radius.

It is recommended that a block of this design be used **only** for K38 standby generator set applications or those applications requiring a rated power of 675 kw [900 bhp] or less.

Note : This section applies to All Cylinder Blocks.

⚠ CAUTION ⚠

Make sure the micrometer contacts the flat surface of the ledge. It must not touch the radius.

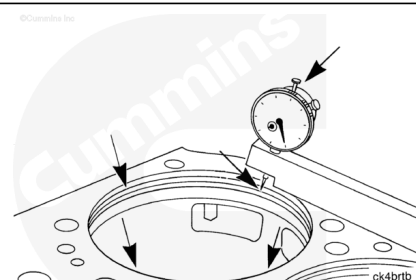


Use a depth micrometer. Measure the counterbore depth at the four places shown.

The four measurements **must not** vary more than 0.25 mm [0.001 in]. If the measurements exceed the specification, the counterbore ledge **must** be machined. Refer to Procedure 001-058 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html>)

⚠ CAUTION ⚠

Be sure that the indicator does not contact the counterbore radius on a block that does not have a double undercut.

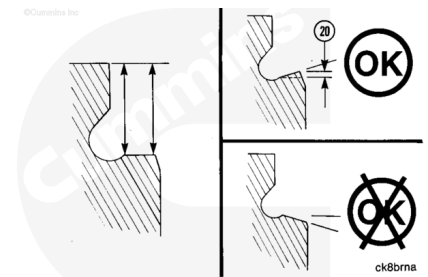


Use an indicator and a gauge block, Part Number 3164438, or equivalent. Measure the angle of the counterbore ledge at no less than four places on the counterbore circumference.

The measurement of the ledge depth **must** be as near to the counterbore radius as possible, and as near to the counterbore edge as possible.

The angle (20) of the counterbore ledge is acceptable (OK) if the measurement that is near the counterbore edge is the same, or no more than 0.036 mm [0.0014 in] shorter, than the measurement near the counterbore radius.

If the measurement near the counterbore ledge is longer than the measurement near the counterbore radius, the ledge **must** be machined. Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

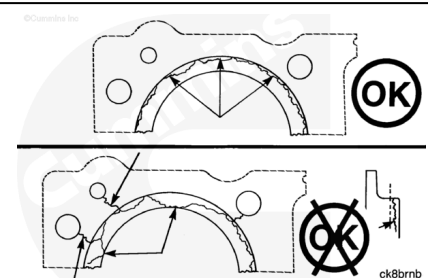


Use a Crack Detection Kit, Part Number 3375432, or equivalent. Check the counterbore ledge for cracks.

Circumferential cracks are acceptable (OK) if they do **not** extend more than half of the distance across the ledge.

Any cracks that extend into a water hole are **not** acceptable (**not** OK). Do **not** use the block.

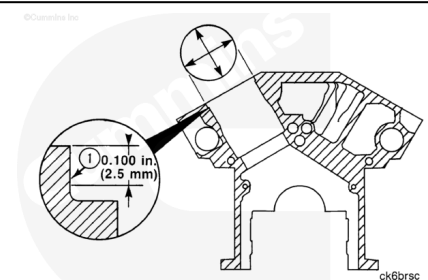
Any cracks that extend into a capscrew hole are **not** acceptable (**not** OK). The block can be used if a capscrew hole thread insert is installed in the capscrew hole. See Alternative Repair Manual, Bulletin 3379035.



Measure the inside diameter of the upper counterbore. The point of measurement **must** be within 2.5 mm [0.100 in] from the top of the block (1).

The inside diameter of the upper counterbore **must** be round within 0.025 mm [0.001 in]. If the measurement is **not** within specification, check to determine if the block can be machined for oversize liners. Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

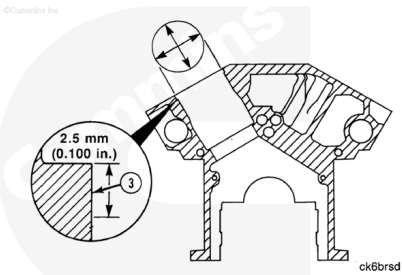
Note : The upper counterbore **must** be no more than 0.025 mm [0.001 in] larger than the cylinder liner flange, or no more than 0.076 mm [0.003 in] smaller than the cylinder liner flange.



Measure the inside diameter of the lower counterbore. The point of measurement **must** be within 2.5 mm [0.100 in] from the top of the counterbore ledge (3).

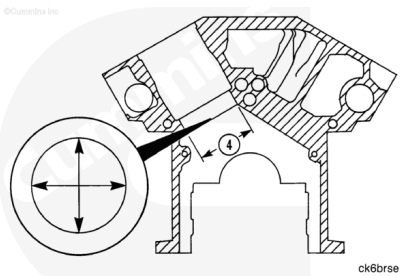
The inside diameter of the lower counterbore **must** be round within 0.025 mm [0.001 in].

If the inside diameter of the lower counterbore is larger than the specification, the block can be used **only** if the outside diameter of the liner flange is 0.025 mm [0.001 in] larger than the inside diameter of the upper counterbore (1). If there is **not** a press fit between the liner and the upper counterbore, the block **must** be machined for oversize liners.



Check the chamfer at the top of the packing ring bore. Excessive pitting **must** be repaired. See the Alternative Repair Manual, Bulletin 3379035.

Measure the Inside Diameter.

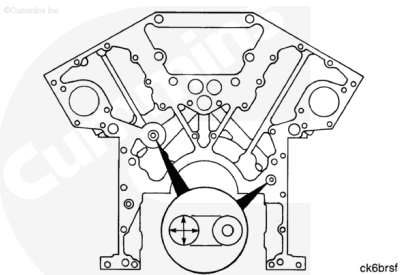


(4) Packing Ring Bore

mm		in
177.32	MIN	6.981
177.48	MAX	6.987

Measure the inside diameter of the bore for the camshaft idler shafts.

Note : Some engines have camshaft idler shaft bores that have the same inside diameter specification as the new water pump idler bore.



Camshaft Idler Shaft Bore Inside Diameter

mm		in
25.362	MIN	0.9985
25.412	MAX	1.000

If the bore is **not** within specifications, a repair sleeve can be installed. See to the Alternative Repair Manual, Bulletin 3379035.

The bore for the water pump idler shaft can be one of two designs.

Newer cylinder blocks and blocks that have been repaired for the new shaft have a counterbore (13) and (14).

Dimensions **must** be within 0.051 mm [.002 in] at all locations on any bore.

Water Pump Idler Shaft Counter Bore - New (13)

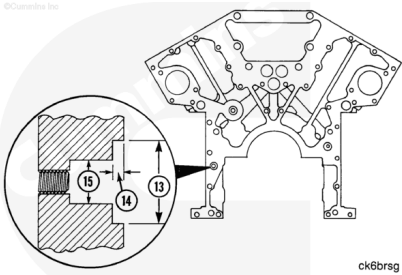
mm		in
43.18	MIN	1.7000
43.244	MAX	1.7025

Water Pump Idler Shaft Counter Bore Depth - New (14)

mm		in
4.57	MIN	0.180
5.08	MAX	0.200

Water Pump Idler Shaft Bore - New (15)

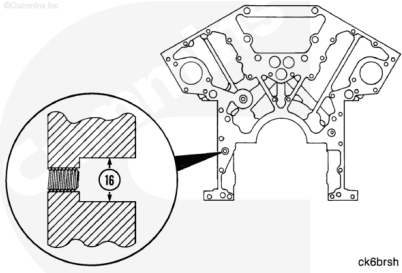
mm		in
22.00	MIN	0.866
22.25	MAX	0.876



Older cylinder blocks do **not** contain the counterbore.

Water Pump Idler Shaft Bore - Old (16)

mm		in
25.37	MIN	0.999
25.40	MAX	1.000



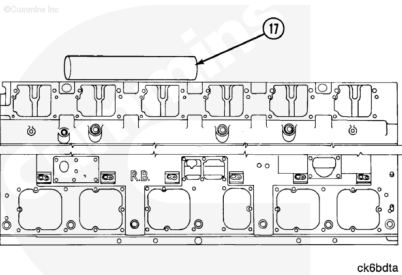
An Idler Bore Kit, Part Number 3822548, is available to modify the bore for the new shaft. The old shaft is available for service.

Use a precision straight edge, Part Number 4918219, or equivalent (17).

Check the head deck of the block for wear. If fretting is visible in an area where a head gasket seal ring or a grommet touches, the surface **must** be repaired.

Fretting damage in any other area is acceptable if it does **not** reduce the ability to measure the liner protrusion accurately.

See the Alternative Repair Manual, Bulletin 3379035, for machining instructions.

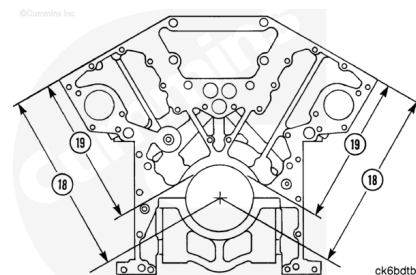


A newly machined surface **must** be flat within 0.051 mm [0.002 in] under a cylinder head. Waves on the surface are acceptable as long as they are no more than 0.018 mm [0.0007 in] in height. The high and low points of the waves **must not** be any closer than 25.4 mm [1.0 in].

A newly machined surface **must** be within the specifications for block height as stated in the next illustration. If the top surface of the block is machined, the ledge depth of the cylinder liner counterbore **must** be machined. Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

⚠ CAUTION ⚠

The parting line of the main bearing cap is not the same height as the centerline of the main bearing bore.



Measure the height of the cylinder block.

(18) Cylinder Block Height

mm		in
481.94	MIN	18.974*
482.78	MAX	19.007

* If the height of the block is less than 482.45 mm [18.994 in], a 0.51 mm [0.020 in] oversize head gasket **must** be installed.

If the checking ring or the centering ring is **not** available, the height of the block can be measured from the top of the main bearing saddle.

(19) Cylinder Block Height - Main Bearing Saddle

mm		in
395.00	MIN	15.551*
395.81	MAX	15.583

* If the block height (19) is less than 395.48 mm [15.570 in], a 0.51 mm [0.020 in] oversize head gasket **must** be installed.

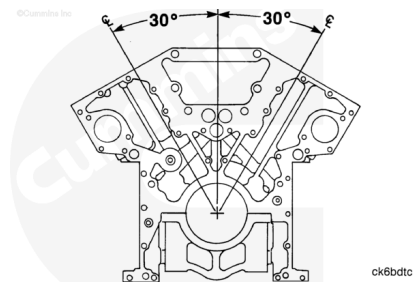
The height of the block **must not** vary more than 0.08 mm [0.003 in] from end-to-end of the block. If the block height is **not** within specifications, the top surface of the block **must** be machined or the block **must** be replaced.

Note : If the head deck surface of the block is machined, the ledge depth of the cylinder liner

counterbore **must** be machined. Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

The cylinder block height of the two engine banks **must not** vary more than 0.13 mm [0.005 in] between the banks.

The cylinder bore angle does **not** need to be measured, but **must** be maintained if the cylinder head surface is machined.



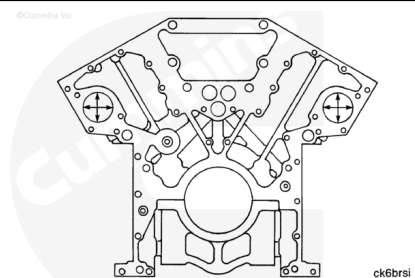
Check the amount of copper exposure on the camshaft bushing. If it extends more than 120 degrees around the bushing, the bushing **must** be replaced. Refer to Procedure 001-010 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-010-tr.html)

Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

Measure the inside diameter



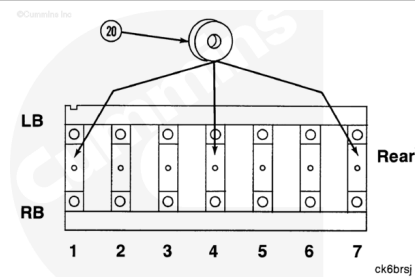
Camshaft Bushing Inside Diameter		
mm		in
76.200	MIN	3.0000
76.289	MAX	3.0035

If the bushing is **not** within specifications, the bushing **must** be replaced. Refer to Procedure 001-058 in Section 1.

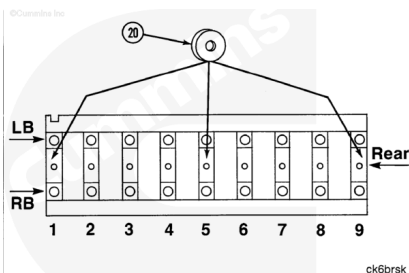
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

Measure Main Bearing Bore Alignment

Use three centering rings, Part Number 3375063, 17.368 mm [6.8465 in] diameter. Put the rings in the number "1", "4", and "7" main bearing saddles of the K38 cylinder block.

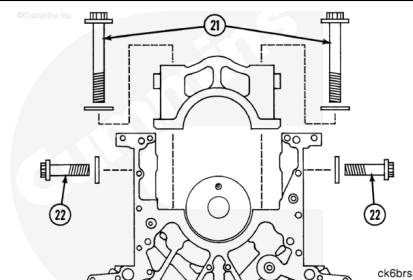


On the K50 engine, put the rings in the number "1", "5", and "9" main bearing saddles.



⚠ CAUTION ⚠

The main bearing caps are numbered through "7" on the K38, and through "9" on the K50 engines. Install each cap in the correct location. Number "1" cap is to be installed at the front of the block. The slot in the cap for the bearing tang must be on the same side as the slot in the block.



Use clean engine oil to lubricate the capscrew threads, heads, and washers.

Install each capscrew and washer (21) by hand to be sure they are positioned properly.

Use a mallet to install the caps until they touch the block.

If any of the caps do **not** require force during installation, mark the cap to check the side clearance.

See Alternative Repair Manual, Bulletin 3379035, for instructions to replace the cap.

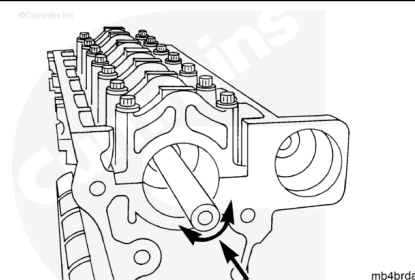
Install and tighten the side bolts (22) and washers. Refer to Procedure 001-006 in Section 1, for side bolt and main bearing capscrew torque steps.

(/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)

Note : The next three illustrations cover checking the number "1" main bearing bore on a K19 engine. The procedure and specifications are the same for the K38 and K50 engines.

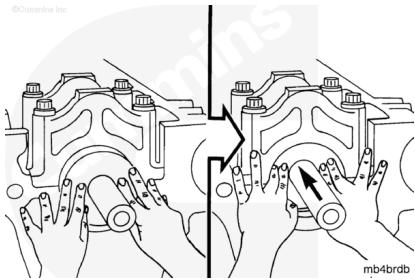
Use clean engine oil to lubricate the inside diameter of the centering rings.

Install the alignment and boring bar, Part Number 3375061, in the centering rings. The bar **must** turn easily. If the bar does **not** turn easily, check to make sure that the main bearing caps are installed correctly. If they appear to be correct, move one of the centering rings to another bearing location.



Install the checking ring, Part Number 3375064, 173.81 mm [6.8430 in] diameter, by hand, in the main bearing bore. If the ring does **not** slide through the bore, check the bore for burrs. If the ring still does **not** slide through the bore, the bore is undersize and **must** be repaired.

See the Alternative Repair Manual, Bulletin 3379035.



Use a 0.076 mm [0.003 in] feeler gauge that is a maximum of 13 mm [0.5 in] wide.

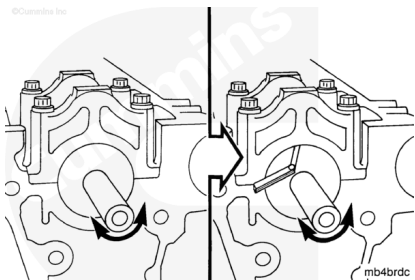
Center the checking ring in the bore. Try to put the feeler gauge between the checking ring and the bore. Rotate the gauge in the bore at both sides of the checking ring.

The bore alignment of the main bearing is acceptable if:

- the gauge does **not** enter at any point
- the gauge will enter, but will **not** slide through or around the bore, and the alignment bar will rotate with the gauge inserted.

The bore alignment of the main bearing is **not** acceptable if:

- the gauge enters and slides around the bore. This means that the bore is oversize and **must** be repaired.
- the gauge will enter on one side **only**, but can slide around the bore. This means that the bore is tapered and **must** be repaired.



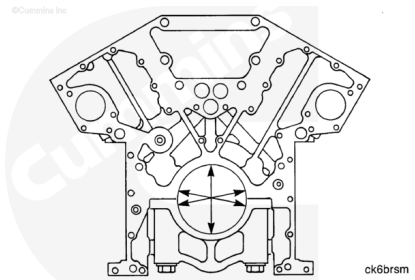
If the tools to check the main bearing bore alignment are **not** available, use a dial bore indicator.

Measure the inside diameter in the three positions shown. The inside diameter **must** be round within 0.013 mm [0.0005 in].

Main Bearing Bore Inside Diameter (Capscrews Torqued to Specification)

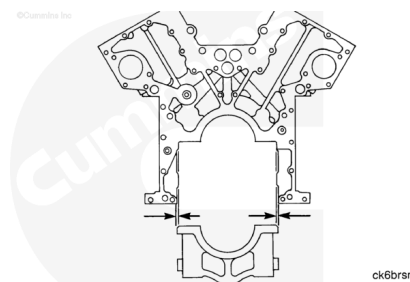
mm		in
173.86	MIN	6.845
173.89	MAX	6.846

Note : Make sure the weight of the dial bore indicator does **not** interfere with the measurement. Keep the solid stem of the indicator against the wall of the bore.



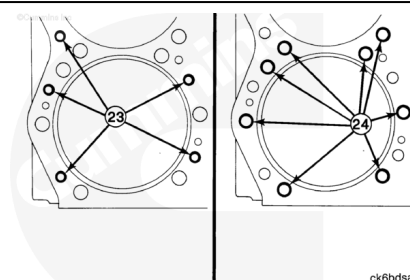
Check the main bearing caps that are loose. The main bearing cap **must** be replaced if it is **not** 0.000 to 0.025 mm [0.0000 to 0.0010 in] larger than the block as shown.

Service caps do **not** have the bore machined to a final specification. If a cap is replaced, the main bearing bore **must** be machined. Use the correct parts of the main bearing boring tool, Part Number ST-1177. See the Alternative Repair Manual, Bulletin 3379035.



Check the water holes (23). If erosion or pitting exceeds 0.08 mm [0.003 in] deep, or extends more than 2.41 mm [0.095 in] from the edge of the hole, the water hole **must** be repaired. See the Alternative Repair Manual, Bulletin 3379035.

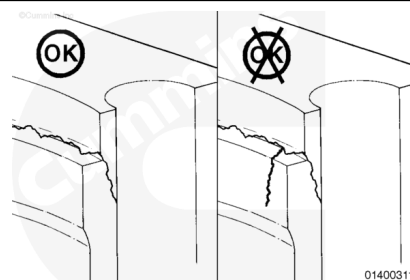
Check the threads of the bolt holes (24) for damage. Use a threaded insert if a damaged bolt hole **must** be repaired. See the Alternative Repair Manual, Bulletin 3379035.



Magnetic Crack Inspect

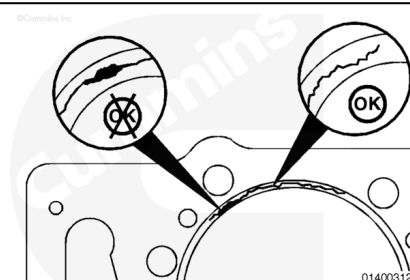
Use a crack Detection Kit, Part Number 3375432, or equivalent, to check the counterbore ledge for cracks.

Circumferential cracks of the counterbore ledge are acceptable if the cracks do **not** extend to or over the edge of the ledge as shown. Circumferential cracks in the radius are acceptable if they do **not** extend more than 90 degrees around the circumference of the counterbore radius.



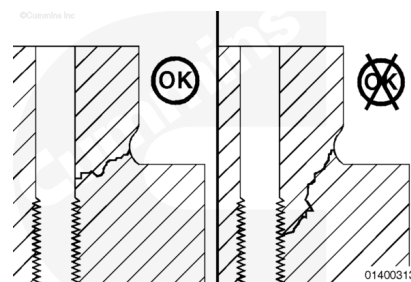
It is **not** necessary to machine the block in an effort to remove acceptable cracks. If cracks that are **not** acceptable are found during the initial inspection, the counterbore ledge **must** be machined. Contact a Cummins® Authorized Repair Location.

If a crack that is **not** acceptable remains after the machining repair procedure is completed, the block is **not** acceptable for reuse.



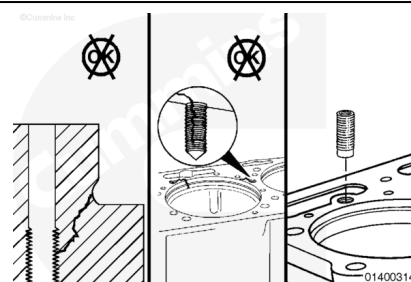
Check the capscrew holes for cracks.

Cracks that extend from the counterbore wall to the capscrew hole are acceptable for use **only** if they do **not** extend into the threaded portion of the hole.

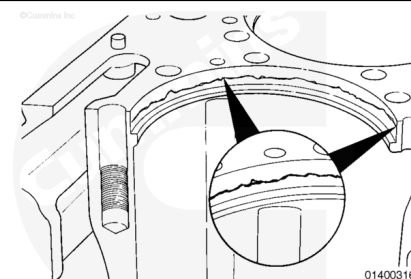


Cracks that extend into the threaded portion of the hole require repair with a blind-end thread insert.

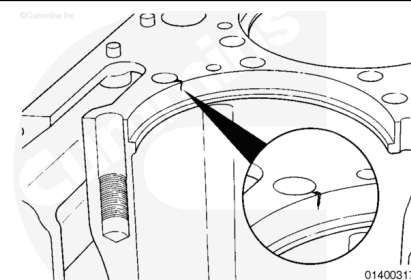
Contact a Cummins® Authorized Repair Location for the proper threaded inserts.



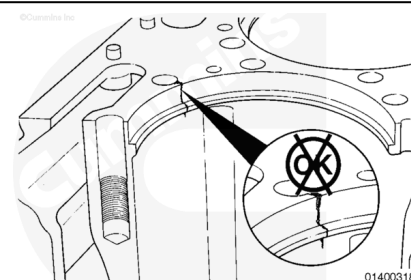
Check for cracks running horizontally around the vertical wall of the counterbore. All coolant passages that are close to the bore **must** be repaired with coolant passage threaded inserts.



Check for cracks that run vertically to a coolant passage or a capscrew hole. Those passages **must** be repaired with coolant passage threaded inserts.

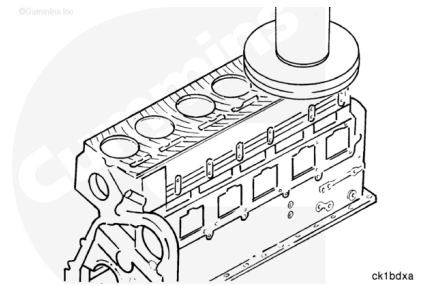


Cylinder blocks with vertical cracks that extend from a coolant passage down over the counterbore ledge can **not** be repaired.

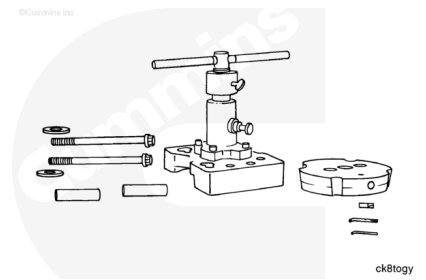


Machine

Resurface the cylinder block, if required, before machining the counterbore ledge depth.

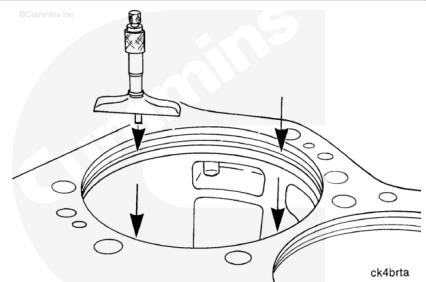


Use the cylinder block counterbore tool, Part Number 3376563, to machine the counterbore ledge depth.



Use a depth micrometer to measure the counterbore ledge depth in the four locations illustrated.

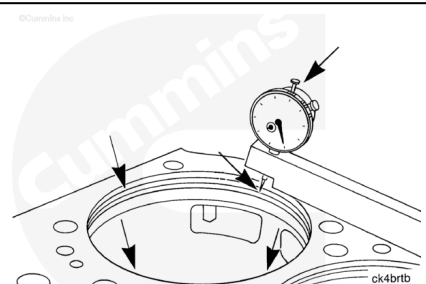
The measurements **must not** vary more than 0.25 mm [0.001 in]. If the measurement exceeds the specification, the counterbore **must** be machined again.



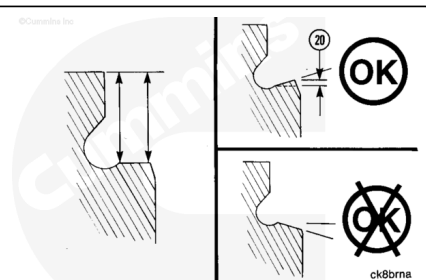
The indicator **must not** touch the counterbore radius on a block that does **not** have a double undercut.

Use the indicator and gauge block, Part Number 3164438, or equivalent, to measure the angle of the counterbore ledge at four equally spaced locations.

The measurement of the ledge depth **must** be performed as near to the counterbore radius as possible, and as near to the counterbore ledge edge as possible.



The angle (20) of the counterbore ledge is acceptable (OK) if the measurement that is near the counterbore edge is the same or no more than 0.036 mm [0.0014 in] shorter than the measurement near the counterbore radius.



If the measurement near the counterbore ledge is longer than the measurement near the counterbore radius, the ledge **must** be machined. Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

⚠ CAUTION ⚠

Concentricity must be within specifications. Engine damage will result when the liner is not seated correctly in the bore.

Check the cylinder block liner bore concentricity. Refer to Procedure 001-070 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-070-tr.html)

The bore **must** be repaired when consistent readings are **not** obtained. See the Alternative Repair Manual, Bulletin 3379035.

